Unit 4 Solutions and Solubility Test - Student Version (verA)

Subject: chemistry
Course Code: SCH3U
Unit: Unit 4 Solutions and Solubility Test

Instructions: Answer all questions. Show full work where required. Write clearly and use appropriate notation.

Written Response

Section Plan: 5 questions | 20 marks

Instructions: Answer all questions. Show all work for calculations, include units, and use appropriate significant digits.

1. A window-cleaning solution is 18.5% (v/v) ammonia. If a bottle contains 750 mL of the solution, what volume of pure ammonia is present? (3 marks)

2. A solution is prepared by dissolving 0.180 mol of sodium carbonate, $Na_2CO_3(s)$, to make 1.50 L of solution. What is the molar concentration of the solution? (3 marks)

3. A stock solution of 12.0 mol/L hydrochloric acid, HCl(aq), is available. What volume of stock solution is needed to prepare 500.0 mL of 1.50 mol/L HCl(aq)? Describe how you would prepare the solution. (5 marks)

4. If 18.0 mL of KOH(aq) is required to completely neutralize 25.0 mL of 0.150 mol/L $H_2SO_4(aq)$, what is the concentration of the base? Write the balanced chemical equation, include states, then solve. (5 marks)

5. Describe one specific example of how water being a good solvent is an advantage and one specific example of how it can be a disadvantage. Explain each example using chemistry ideas. (4 marks)